



Invited review: Rumen modifiers in today's dairy rations

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ABSTRACT

Our aim was to review feed additives that have a potential ruminal mechanism of action when fed to dairy cattle. We discuss how additives can influence ruminal fermentation stoichiometry through electron transfer mechanisms, particularly the production and usage of dihydrogen. Lactate accumulation should be avoided, especially when acidogenic conditions suppress ruminal neutral detergent fiber digestibility or lead to subclinical acidosis. Yeast products and other probiotics are purported to influence lactate uptake, but growing evidence also supports that yeast products influence expression of gut epithelial genes promoting barrier function and resulting inflammatory responses by the host to various stresses. We also have summarized methane-suppressing additives for potential usage in dairy rations. We focused on those with potential to decrease methane production without decreasing fiber digestibility or milk production. We identified some mitigating factors that need to be addressed more fully in future research. Growth factors such as branched-chain volatile fatty acids also are part of crucial cross-feeding among groups of microbes, particularly to optimize fiber digestibility in the rumen. Our developments of mechanisms of action for various rumen-active modifiers should help nutrition advisors anticipate when a benefit in field conditions is more likely.

Key words: rumen modifiers, lactate metabolism, methanogenesis

INTRODUCTION

Increasing starch in dairy rations to provide energy for milk production simultaneously increases supply of rumen-degraded starch to ruminal microbes. Although likely increasing microbial protein synthesis, corresponding decreases in efficiency of microbial protein synthesis, NDF degradability, or milk fat percentage

dampen the benefit from increased starch (Ferraretto et al., 2013; Hackmann and Firkins, 2015; Dewanckele et al., 2020; Firkins, 2021). Negative associative effects can be minimized through practices such as monitoring forage or TMR particle size, decreasing sorting behavior, and balancing starch fermentability as affected by grain processing. However, after these managerial strategies are adopted on farms, rumen modifiers can be considered for a role in dairy diets to help optimize fiber degradation and efficiency of microbial protein synthesis to use nutrients more efficiently and thereby lessen environmental impact.

Further increases in starch can increase the risk of SARA. Considerable knowledge has been derived from SARA challenges, particularly for the epimural (those that interact with the gut epithelium) microbes in the rumen and hindgut (Plaizier et al., 2022). However, there is relatively little known about how rumen modifiers can be strategically used to influence rumen fermentation in moderate to moderately high starch diets that do not induce SARA or only do so periodically or with a fraction of cows.

Because there is so much variation among animals in methane intensity (methane produced per unit of ECM) and feed efficiency (Firkins, 2021), many researchers have questioned whether or not the rumen microbiome can be manipulated, including with usage of dietary modifiers. We often must extrapolate between microbiome 16S rRNA gene sequence analyses from samples collected *in vivo*, which often under-represent the adherent cellulolytics because the particulate matter was not recovered (Firkins and Yu, 2015). Many of the conditions establishing a pure culture can involve a substrate or other type of shock that could mimic an abrupt shift to a high grain diet (Tajima et al., 2000), whereas substrate types and concentrations *in vitro* are not always physiological (Hackmann and Firkins, 2015), particularly to assess fiber degradation. Many of the inferences from various rumen modifiers have been extrapolated from these ranges in conditions; hence, our premise is that a representation of mechanism will help expand their generalizability for when and when not to be used. Several rumen-active feed additives have been described (NASEM, 2021), but mechanism was not expanded thoroughly.

Received August 11, 2022.

Accepted November 23, 2022.

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Many processes in ruminal metabolism involve multiple steps and multiple microbial taxa. A premise to understanding these steps starts with avoiding “fermentation” as a catch-all word for everything that happens in the rumen. Carbohydrates must be degraded primarily by adherent fibrolytic or amylolytic bacteria before the sugars being transported inside cells (Moraïs and Mizrahi, 2019). Partially degraded starch granules or fibrous particles laden with bacteria can be engulfed inside protozoal cells (Firkins et al., 2020), although most research with rumen modifiers has attempted to inhibit protozoa and will not be described herein. Few recent studies have used modifiers to stimulate or inhibit specific taxa involved in biohydrogenation of UFA, so this process will not be discussed. The AA available from RDP can be fermented or can undergo Stickland reactions to produce ATP. Some additives have potential to improve efficiency of microbial protein synthesis by optimizing the ratio of catabolism versus anabolism of sugars from degraded starch and fiber into more efficiently growing cells (Russell and Cook, 1995). The role of thermodynamics of VFA to control VFA production has only marginal control (Li et al., 2022), probably because there presumably was no change in cellular redox. Instead, our primary aim will be to explain how reducing equivalents are produced and used in cell growth to direct how, when, and why microbes produce different end products, some of which are substrates for other microbes. Our objectives are to translate the scientific literature to provide context for explaining modes of action for rumen modifiers that are being considered for use alone or in combination by researchers and dairy nutrition advisors.

ADDITIVES TO MANIPULATE RUMEN FERMENTATION

Review of Electron Transfer in Fermentation

Published fermentation pathways typically show flow of metabolites, whereas fermentation pathways are directed by gene expression of the microbial community members and the flow of electrons down those expressed pathways. Pathway flux can be interrupted by allosteric modifiers, phosphorylation, and various repressors or activators. These types of mechanisms are beyond our scope. In contrast with control by such regulation, flux through fermentation pathways also involves thermodynamics of reactants and coupled cofactors. Typically, the electrons accepted by reduced enzyme cofactors are “sunked” as part of the fermentation end product itself, especially propionate, valerate, or caproate. The recycled (oxidized) cofactor can then be used again when another sugar is fermented. The

cycle must continue for prolonged VFA production and optimum microbial growth.

There typically are pairs of electrons with corresponding pairs of protons transferred by cofactors such as $\text{NAD}^+ \leftrightarrow \text{NADH} + \text{H}^+$, $\text{NADP}^+ \leftrightarrow \text{NADPH} + \text{H}^+$, or $\text{FAD}^+ \leftrightarrow \text{FADH} + \text{H}^+$. Collectively, the pairs of electrons and corresponding protons can be abbreviated as $[2\text{H}]$, with the brackets generalizing among different electron carriers even though the cofactors are typically specific to their respective enzymes. Anaerobes, including those in the rumen, use ferredoxin and flavodoxin as electron carriers in key intermediate reactions that reduce the thermodynamic change between reactants and conserve some of the change in free energy to form ion gradients (primarily H^+ and Na^+). Some schemes depict $[\text{H}]$ as a single hydrogen (Beauchemin et al., 2020), but the general depiction of even numbers (e.g., $2 [\text{H}]$) still represents the same relationship as $[2\text{H}]$.

Interspecies hydrogen transfer ripples throughout the entire rumen microbial community. Some of the $[2\text{H}]$ generated by one microbe is released as dihydrogen (i.e., H_2). Contrary to many reports, $[2\text{H}]$ and H_2 are not necessarily interchangeable. About two-thirds of microbes in the rumen have a hydrogenase enzyme that converts $[2\text{H}]$ to H_2 or the reverse reaction (Greening et al., 2019). These hydrogenases can be reversible but tend to be either H_2 -producing or H_2 -using on a net basis, depending on the specific niche of the microbes. Methanogenesis involves 3 main routes, including the 2 most common in the rumen: methane from H_2 or formate (i.e., hydrogenotrophic; formate is readily converted to H_2 by most of these archaea) or from methyl groups (methylotrophic), with only a relatively few in both niches. Moreover, hydrogenases are differentially expressed, with growing evidence that the concentration of dissolved H_2 and its entry inside cells is being sensed by specific hydrogenases for subsequent differential expression of other hydrogenases. Because hydrogenases in hydrogenotrophic microbes, particularly methanogenic archaea, have such a high affinity for H_2 , it is eructated or respired (after absorption into blood) only when methanogenesis is inhibited. Particularly in such cases, if H_2 is produced faster than it is consumed (primarily to produce CH_4), the concentration of aqueous H_2 [hereafter referred to as $\text{H}_2(\text{aq})$] increases and can thermodynamically inhibit H_2 production, and $\text{H}_2(\text{aq})$ can even increase beyond its ability to be dissolved (i.e., is supersaturated). Further details are presented by Ungerfeld (2020). Because alternative $[2\text{H}]$ sinks were proposed to be widely available (Greening et al., 2019), further research needs to explain and help prevent accumulation and expiration of H_2 with methane suppression.

Unique enzymes rely on electron flow at key sites in fermentation pathways (Ungerfeld, 2020). These enzymes are becoming increasingly detected in the genomes of ruminal microbes as a means to conserve energy; typically, these protein complexes combine carbon flux with generation of a H^+ or Na^+ gradient (i.e., higher ion concentration outside than inside cells) that can be used to drive active transport or to produce ATP unless disrupted by an additive such as monensin. More recently described, the electron pairs can be split during bifurcation; the reverse process is termed confurcation (Buckel and Thauer, 2018). One key site to conserve energy involves the Rnf complex (i.e., ferredoxin- NAD^+ oxidoreductase). It couples Na^+ pumped outside the cell to form a gradient that is subsequently used to harness ATP in gut anaerobes via a Na^+ ATPase (Hackmann et al., 2017). Although electron transfer phosphorylation was projected to be significant (Dawson et al., 1979), classical fermentation yields of ATP consistently ignored these routes in favor of substrate-level phosphorylation (i.e., ATP generated as part of the enzymatic reaction). Energy-conserving enzymes particularly involve succinate (Hackmann et al., 2017) and (combined with acetate uptake) lactate (Sheridan et al., 2022) as intermediates; these electron sinks typically compete with the production of methane and are tied to numerous additives' modes of action.

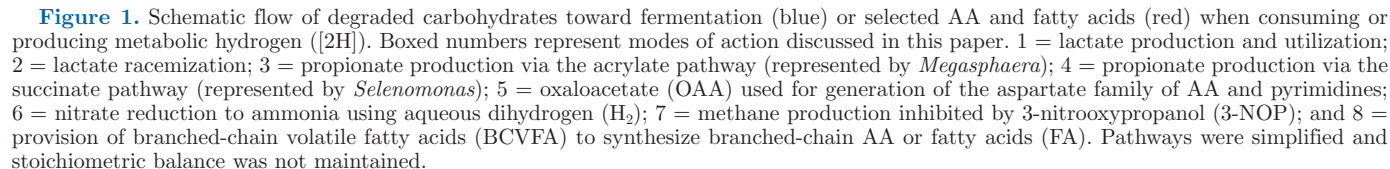
The various reactions are collapsed in Figure 1 to show the importance of $[2H]$ in cellular metabolism (enzymes using electron bifurcation or confurcation intermediate steps are left off for simplicity). Most of the archaea in the rumen are characterized as hydrogenotrophic (using H_2), although much more diversity among methanogens should be expected with forages containing a lot of soluble fiber (Greening et al., 2019). Aqueous H_2 is likely increasingly replaced by formate when methanogenesis is inhibited and $H_2(aq)$ builds up (Kelly et al., 2022). This form of interspecies hydrogen transfer using $H_2(aq)$ or formate is deeply embedded in the entire rumen microbial ecosystem. The $H_2(aq)$ also can be used by some other microbes to produce propionate via fumarate reduction; the importance of this process is not well known but perhaps is greater than previously thought (Greening et al., 2019). Those researchers also have assessed the potential for homoacetogenesis [acetate produced from $H_2(aq)$ and dissolved CO_2] in the rumen, although conditions might be more favorable for acetogens in the hindgut than the rumen (Firkins and Yu, 2015). Alternate electron sinks such as nitrate or sulfate have been fed in studies to compete with methanogens for $H_2(aq)$, particularly nitrate (Yang et al., 2016). In this case, $[2H]$ derived from fermentation is coupled with nitrate reductase and nitrite reductase to ultimately produce NH_3 , the main

N source for rumen bacteria. These reductase products are "alternate" to producing H_2 , which would otherwise be consumed in methanogenesis.

Figure 1 and the following discussion is patterned after that of Ungerfeld (2020). However, readers need to view this and all such figures as a set of reactions that are summative over a community of microbes each of which has its own genome and differential expression of that genome with respect to these fermentation pathways. Acetate pathways are widely expressed to make ATP, so acetate is produced about 3 times more than propionate in the rumen of dairy cattle. Because the acetate pathway generates more $[2H]$ than is used, that extra $[2H]$ must be sunk to some other reduced end product to allow reoxidation of cofactors for continued fermentation. As stated previously, many fibrolytic bacteria encode genes to produce acetate to produce ATP and either H_2 or formate (which can be converted to $H_2 + CO_2$) to recycle electron carriers. Sinks for $[2H]$ include longer-chain VFA such as butyrate, valerate, and caproate compared with acetate. Moreover, increased cell growth provides a natural $[2H]$ sink in cellular products such as AA and fatty acids. More rapidly growing cells therefore naturally sink more $[2H]$ in their cellular constituents compared with slower growing microbes. If $[2H]$ produced in fermentation builds up beyond what can be used for cell growth, though, dehydrogenase and hydrogenase enzymes would be thermodynamically stalled. Several founding studies documented that fiber degradation by cellulolytics decreased unless they were cocultured with a secondary microbe that consumes the cellulolytic's $[2H]$ sink (Mizrahi et al., 2021). For example, if succinate or H_2 are released and not used by noncellulolytics, the cellulolytics would simultaneously accumulate $[2H]$ intracellularly and thermodynamically slow fermentation, ATP generation, and ultimately cellulolysis. Many additives discussed subsequently might help prevent these stalls in $[2H]$ cycling.

Lactate as an Intermediate

Most ruminant nutritionists would recall tissue lactate dehydrogenase (**LDH**), particularly for the reaction catalyzing the L-stereoisomer of lactate (**L-LDH**): pyruvate + $NADH + H^+ \leftrightarrow$ L-lactate + NAD^+ (reaction 1 in Figure 1). This reaction could be rewritten as $C_3H_4O_3 + [2H] \rightarrow C_3H_6O_3$ (note the $[2H]$ sinking into with pyruvate to add 2 H atoms in lactate). Because dehydrogenase enzymes have affinities for both metabolites and cofactors on each side of the equation, the reaction's direction and rate depend on their relative concentrations with respect to the specific enzyme's expression. Lactate racemase (see reaction 2 in Figure 1) probably allows more lactate to be produced because



Lactate is a stronger acid than the VFA by about one log (Sheridan et al., 2022). That is, at the same concentration, lactate would drop pH by one whole pH unit (10-fold higher H^+ concentration) compared with the VFA, which all have similar pKa; hence, lactate accumulation has long been associated with SARA. Greater glucose availability with greater starch fermentability

Despite being a stronger acid, lactate can be fed to dairy cows without promoting SARA so long as the lactate is extensively fermented to VFA. Dosed ammonium lactate was catabolized primarily to propionate almost completely within 1.5 h in continuous cultures (Wagner et al., 2018). Because buffers were adjusted to control pH above 6.0, lactate had a minor effect on bacterial populations. In diets that are moderate in rumen-fermentable starch, similar increases in molar proportions of propionate (and butyrate) were detected in cows fed this lactate source (Caputo Oliveira et al.,

2019). Metabolic indicators suggested that supplemental lactate increased liver supply of propionate in transition cows (Caputo Oliveira et al., 2020). Those authors noted improved milk efficiency resulting from decreased DMI while maintaining similar milk production. An improvement in feed efficiency also was noted when dosing a propionate-producing probiotic (Weiss et al., 2008). These responses are consistent with the theory that propionate supply beyond needs for gluconeogenesis increase liver acetyl-CoA oxidation through the citric acid cycle and cause hypophagia (decreased DMI) by dairy cows (Allen, 2020).

Current research is ascertaining how fermentation intermediates (i.e., those produced by one microbe but fermented by another) are related to methanogenesis. Fermentation of propionate via succinate likely involved the genera *Prevotella* and *Selenomonas* in goats (Denman et al., 2015), and fermentation of lactate to propionate and butyrate likely occurred via *Megasphaera* in sheep (Kamke et al., 2016). Both cases were associated with low methane emission. Figure 1 (modes 3 and 4) illustrates how both ways (through succinate or acrylate) of producing propionate consume $[2H]$ that is thereby diverted away from H_2 , thus circumventing methane production. The potential to consume $H_2(aq)$ to produce propionate has been shown in pure cultures and inferred from metatranscriptomics screening (Greening et al., 2019) but awaits further assessment regarding this route. To date, inconsistencies in efficacy of propionate-producing probiotics to suppress methane still need to be worked out (Honan et al., 2021). Even butyrate production yields less H_2 compared with acetate, and many researchers have documented that butyrate fuels the rumen epithelium for absorption or could increase flow of blood draining the rumen to increase rate of diffusion of VFA (Storm et al., 2011). Butyrate supplementation might have value for calves to promote rumen epithelial integrity (Aschenbach et al., 2019). Those authors also highlighted the roles of the lower gut and stressful periods in the etiology of SARA when mature animals are transitioned to high concentrate diets.

Lactate Accumulation

Lactate accumulation will only happen when there is plenty of substrate (i.e., starch that is extensively converted to glucose), particularly when glucose availability exceeds its conversion into metabolic intermediates (often studied by limiting dietary RDP). Lactate yields only 2 ATP/glucose, but lactate production is thought to be at least 5-fold faster than fermentation to VFA (Nagaraja and Titgemeyer, 2007). Thus, lactate producers gain ATP about 2.5-fold faster, and they are

more resilient to the decreasing pH as lactate production rate exceeds its usage rate. There also is some indication that lactate, especially the D-racemer (Nagaraja and Titgemeyer, 2007), is absorbed more slowly from the rumen than the VFA, exacerbating this acidotic situation particularly for those animals that are not adapted (Aschenbach et al., 2019). Those latter authors described the multifaceted nature of SARA in dairy cattle as being more likely than the standard spiral of events leading to acute lactate acidosis, which was described by Jouany (2006).

Rapid and unsustained growth of saccharolytic microbes can be followed by death and then lysis of cell components. The lipopolysaccharide (LPS) components (among other endotoxins) of bacterial cell walls are associated with rumen inflammatory responses (Aschenbach et al., 2019). Zebeli et al. (2012b) noted a threshold for potential systemic inflammation when concentrate exceeds 44% of DM and when diets decreased below 39% NDF, which would occur for many diets for high producing dairy cattle. Increased concentration of LPS appears to exacerbate lactate production without affecting lactate consumption (Dai et al., 2020). Thus, any managerial factors to prevent swings in starch availability should help to keep lactate production from accumulating and resulting in SARA.

Recent research is emphasizing the epithelium (i.e., the epimural) microbial community even under SARA conditions (Petri et al., 2020). The rumen epithelium is microaerophilic because of dissolved O_2 diffusing from the blood, and microbes have adapted to these microaerophilic conditions (Mann et al., 2018). Much more work is needed to understand these epithelium-associated microbes' role in rumen function because they seem to be related to differences in productivity indices such as feed efficiency among animals (Bickhart and Weimer, 2018; Na and Guan, 2022).

Redox and Rumen Ecology

The oxidation-reduction potential (ORP; sometimes called redox potential and abbreviated as E_h) technically measures the tendency of a chemical species to accept or donate electrons to an electrode. It typically has been measured in a similar way as pH using a hand-held device but now can be measured with intraruminal cow monitors (Dijkstra et al., 2020). A lower value (more reduced rumen environment) indicates a lower affinity for electrons (i.e., higher $NADH/NAD^+$), and a higher value (more oxidized; i.e., lower $NADH/NAD^+$ inside cells) indicates a higher affinity for electrons, as those authors visualized for reactions with differing ORP. An even more detailed description of the chemistry is in Ungerfeld (2020).

A more negative ORP favors production of H_2 (Kelly et al., 2022) and increased NDF digestibility (Huang et al., 2018), although the degree to which these associations are mechanistic is not well known. By sensing intracellular H_2 concentration and probably redox, *Ruminococcus albus* can improve its ATP yield by interspecies H_2 transfer with methanogens when ORP is very negative (Zheng et al., 2014). We presume other mechanisms will be discovered to document a greater importance for highly negative ORP to maximize fiber degradation, which is why H_2 (aq) accumulation should be measured along with ORP in methane abatement studies. Cellulolytic specialists (e.g., *Fibrobacter succinogenes* and *Ruminococcus flavefaciens*) produce significant succinate as their electron sink, whereas *R. flavefaciens* and *R. albus* are more likely to produce formate or H_2 during acetate production when cocultured with methanogens (La Reau and Suen, 2018; Neumann and Suen, 2018). Succinate and H_2 are converted primarily to propionate and methane, respectively (modes 4 and 7 in Figure 1), so some researchers have focused on fostering *F. succinogenes* over the ruminococci. Just as this was studied with inoculation of gnotobiotic lambs (Yeoman et al., 2021), further research is needed to study the native cellulolytics and their webs in young animals during key transition stages (Firkins, 2021).

Another mechanism by which the production and usage of succinate and H_2 seem to be favored under a more reduced environment could be related to a greater NADH/NAD⁺ ratio increasing the yield of reduced products to maintain intracellular redox (Dijkstra et al., 2020) but also to key enzymatic reactions requiring cofactors that are very sensitive to O_2 (Meinecke et al., 1989; Chowdhury et al., 2015; Lu and Imlay, 2021). Increasing O_2 entry with feed and water raises ORP, whereas facultative anaerobes respiring dissolved O_2 make ORP more negative, benefiting the strictly anaerobic cellulolytics. We recommend for researchers attempting to suppress methanogenesis without disrupting normal rumen function to measure ORP more consistently. For example, nitrate reduction competes with methanogens for H_2 (aq) to suppress methanogenesis (Mode 6 of Figure 1). However, nitrate feeding also likely increases ORP (Yang et al., 2016).

Researchers can only measure ORP in rumen fluid (extracellular), but this measurement reflects intracellular redox state of enzymes and the likelihood to direct fermentation pathways (Dijkstra et al., 2020). In Figure 2, we fed continuous culture fermenters the same diet but manipulated pH using buffers (panel A). In panel B, we manipulated 2 forage:concentrate ratios while maintaining very similar pH response curves. Decreasing pH (panel A) inherently makes ORP more positive (higher H^+ concentration is related to the measurement

of ORP); however, increasing concentrate (panel B) also increased ORP independent of pH. The combined effect of increasing concentrate with an associated decrease in pH was a 10 to 15% increase in ORP. The positive association of ORP with propionate (Huang et al., 2018) is expected based on previous discussion. However, higher starch diets increase abundance of amylolytics that produce propionate but also decrease pH. Trying to unbundle these sources of multiple correlation (more starch and lower pH) is difficult experimentally (Broudiscou et al., 2014). Therefore, we recommend for researchers to study the effects of ORP as related to these dietary conditions and perhaps microbial community analysis by using multivariate techniques rather than simple correlation analysis (Firkins, 2021).

In contrast with microbiome analyses with stomach sampling, which likely underrepresents fibrolytics that remain adherent in particles, the measurement of ORP from sensors near the rumen mat versus rumen wall might be particularly important. Mann et al. (2018) noted that those bacteria associated with the epithelium have a high ability to quench O_2 that diffuses across the rumen wall from blood. For cows with SARA, repeated exposure of the rumen epithelium to low pH can wear away the barrier of the membrane and allow further O_2 diffusion and oxidative stress (Na and Guan, 2022). Thus, the already higher susceptibility to a more positive ORP could be exacerbated and help tilt the epithelium to a more inflammatory state when facultative anaerobes that promote inflammation increasingly displace the desirable fermentative bacteria that need a more negative ORP (Friedman et al., 2017). Current research suggests renewed understanding of how some additives might improve function of the epimural bacteria that do not stimulate inflammation (Petri et al., 2020). For example, despite being very sensitive to dissolved O_2 concentration, the prominent gut strict anaerobe *Faecalibacterium prausnitzii* likely uses an extracellular electron shuttle to transfer electrons to O_2 (Khan et al., 2012). If we can further reduce variability associated with measuring ORP, including where it is measured, then maybe it will help explain variability among animals per se.

Probiotics and Organic Acids to Stimulate Lactate Consumption

Megasphaera elsdenii and the lactolytic group of *Selenomonas ruminantium* have been characterized as important lactate consumers based on culture-based information, whereas there likely are emerging lactolytic bacteria such as *Coprococcus catus* that seem to be taking up this niche based on 16S rRNA gene surveys of rumen samples (Mizrahi et al., 2021). A human gut

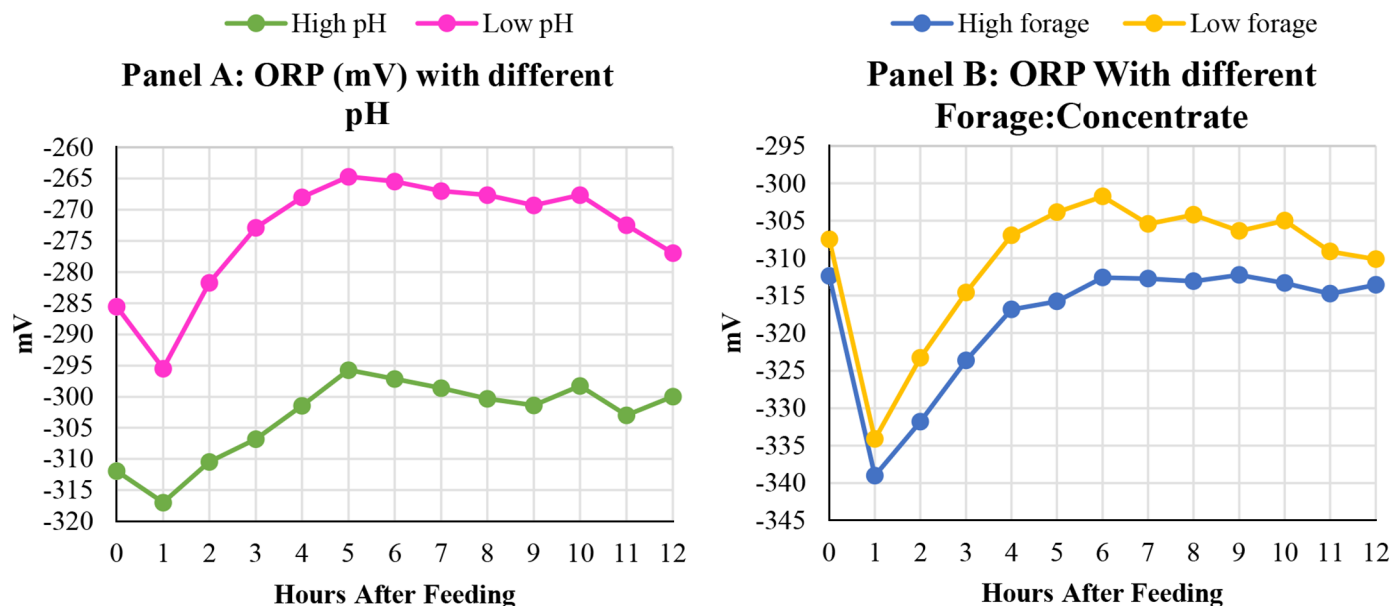


Figure 2. The effects of pH and forage:concentrate, independent from each other, on the oxidation-reduction potential (ORP) in dual-flow continuous cultures. In panel A, a 50:50 forage:concentrate was fed, but buffers were adjusted for pH to range from 6.3 to 6.8 (high pH) versus 5.7 to 6.2 (low pH). In panel B, forage:concentrate ratio ranged from 67:33 (high forage) or 33:67 (low forage), but buffers were adjusted to have very similar pH in the cultures. Data are based on reports by Roman-Garcia et al. (2021b) and Mitchell (2022).

isolate of *C. catus* was documented to produce acetate, propionate (via acrylate), or butyrate from D- or L-lactate under differing conditions (Sheridan et al., 2022). The fermentation scheme in their paper emphasized the importance of a protein complex that couples ferredoxin oxidation with NAD⁺ reduction when lactate was converted to pyruvate in NAD-independent lactate dehydrogenase. Because pyruvate conversion to lactate using NAD-dependent LDH is not energetically favorable, this electron-transferring flavoprotein complex using NAD-independent LDH combined with Rnf (as in *C. catus*) was suggested to be critical to the lactilytic niche by anaerobes (Weghoff et al., 2015). Further metagenomics studies should be associating expressed LDH isozymes from mixed communities with potential for additives to alleviate SARA or inflammation. Until then, *Megasphaera* and *Selenomonas* still are useful culture models to describe the 2 pathways that consume the [2H] by producing propionate through acrylate and succinate, respectively (modes 3 and 4 in Figure 1). Both lactilytics clearly have evolved ways to use electron carriers to produce ATP (Ungerfeld, 2020) and therefore should benefit from highly reduced ruminal environment (more negative ORP) to produce propionate.

Because production of propionate is a reductive process ([2H] is sinked), a lactilytic microbe producing propionate via the succinate pathway would have even more [2H] to dispose of when lactate is converted

to pyruvate. Hence, the lactilytic *Selenomonas* strains improved lactate uptake when cultures were dosed with malate and fumarate (Martin, 1998). That author explained that those dicarboxylic acids can be in feeds such as some forages. However, they also are in feed additives. Finally, these citric acid cycle intermediates can be produced by live yeast and are in yeast extracts (Jouany, 2006). Malate and fumarate have 2 critical roles: (1) providing an electron sink needed to make succinate and (2) providing carbon needed for cell synthetic reactions (see modes 4 and 5 in Figure 1). The critical compound oxaloacetate (mode 5) is at the juncture of propionate synthesis and anabolic pathways for AA and nucleic acids. In addition, because bacteria such as the cellulolytic *Fibrobacter succinogenes* produce succinate, succinolytic *Selenomonas* was projected to aid fiber degradation by alleviating succinate accumulation and thermodynamic inhibition of succinate production (Sawanon et al., 2011). Both *Selenomonas ruminantium* and *Succiniclasticum ruminis* have been described as core members of the succinolytic niche (Mizrahi et al., 2021). In addition to providing organic acids, live yeast (Meller et al., 2019) or yeast extract (Tun et al., 2020) have increased *Bifidobacterium* relative abundance in the rumen of cows. Long associated with milk feeding and interaction with the intestinal mucosa (Kelly et al., 2016), the bifidobacteria might help stabilize the rumen under acidic conditions, perhaps through interacting with

the epimural bacteria in the rumen epithelium (Meller et al., 2019).

Fumarate reductase (producing succinate from fumarate) is a key enzyme needed for propionate producers to make ATP via electron-transport phosphorylation (Hackmann et al., 2017). This enzyme's activity decreased as pH declined below 6.0 (Asanuma and Hino, 2000). Substrate-level control might explain why Martin (1998) inferred that providing malate or fumarate increased in importance as pH declined below 6.0. Supplementing fumarate might prevent downswings in ruminal pH (which increases ORP), should help improve efficiency of lactate fermentation to propionate, and therefore decrease inhibition by accumulated lactate. A more negative ORP from yeast was associated with more lactate consumption in dairy cows (Marden et al., 2008).

Megasphaera elsdenii has been studied as a probiotic based on numerous results from research using culture-based approaches. There has been some conflicting discussion whether or not increasing *M. elsdenii* abundance increases *trans*-10,*cis*-12 CLA concentration or if this lactolytic bacterium increases concomitantly with conditions that predispose cows to milk fat depression caused by other bacteria that produce this CLA isomer. Most likely, *M. elsdenii* does not contribute significantly in a direct way to milk fat depression (Dewanckele et al., 2020). In 2 dairy studies with relatively high amounts of starch (30 and 31%) that was provided by highly available sources (wheat and barley), 2 rather different trends were detected. When DMI numerically decreased by about 2 kg/d with *M. elsdenii* dosing, propionate concentration in the rumen was increased (Aikman et al., 2011). When DMI was not affected by *M. elsdenii* dosing, propionate was unchanged, but butyrate increased (Zebeli et al., 2012c). This latter study had much more NDF and forage NDF than the previous study. In general, *M. elsdenii* is thought to produce mostly propionate from lactate, but it produces more butyrate from glucose if lactate decreases (Weimer and Moen, 2013). Allen (2020) has described how increased propionate delivery to the liver can influence feeding behavior and lower DMI; it also is insulinogenic, potentially partitioning energy to adipose instead of mammary tissues. Thus, these types of very high rumen-available starch situations with less NDF might be riskier for this additive because of the potential for depressing DMI than for depressing milkfat. In studies with moderate dietary starch, those animals that were more likely to respond (i.e., high producing and multiparous) were more likely to benefit from dosed *M. elsdenii* (Stevens et al., 2017; Ye and Eastridge, 2018).

Yeast Products

Live yeast and yeast extract have found their places in many dairy rations to help optimize fiber degradation and improve DMI (Adesogan et al., 2019). We note that fungal extract appears to have some of the same properties as yeast extract, but we could not find adequate peer-reviewed summaries related to ruminal function to include in this report. Jouany (2006) suggested oxygen scavenging by live yeast that congregated to newly ingested feed (which includes trapped atmospheric O₂) benefited those fibrolytic bacteria colonizing newly ingested feed. Oxygen scavenging was supported as a major role for live yeasts and potential boosting of cellulolytic bacterial abundance (Pinloche et al., 2013) and probably for pH control via increased lactate uptake (Marden et al., 2008). Although isotrichid (and perhaps other) protozoa and facultative anaerobic bacteria also can help remove O₂, live yeast would be more likely to respire O₂ beyond basal microbial responses when compared with yeast culture. Newbold et al. (1996) discussed relatively little benefit expected from malate derived from yeast, although their continuous cultures' pH was buffered near 7.0 (which might have limited the response; see earlier discussion). Even extract that was experimentally collected from live yeast stimulated lactate fermentation to help maintain pH (Rossi et al., 1995) or stimulate growth of *Selenomonas*, *Ruminococcus*, and *Fibrobacter* strains (Callaway and Martin, 1997). The latter authors suggested that extract included soluble growth factors for these other microbes. However, the extract also might have some indirect benefit to help keep ORP low. For example, cysteine is a natural O₂ quencher for batch cultures (Broudiscou et al., 2014). Various antioxidants such as glutathione (containing cysteine) are in *Saccharomyces cerevisiae* (Meng et al., 2017).

Live yeast or yeast extract probably have additional roles still being assessed. Petri et al. (2020) suggested that yeast extract stimulated expression of transporters in the rumen epithelium that could increase rate of VFA absorption and thereby increase pH. Yeast extract has improved feeding frequency (DeVries and Chevaux, 2014; Yuan et al., 2015; Dias et al., 2018), although rumination bouts decreased in another study (Longuski et al., 2009). More frequent feeding might be expected if improved NDF degradability decreased the time to return to eating (Olagaray et al., 2019). Those authors noted a potential to improve milk fat synthesis but also potential for subclinical ketosis and BHB production, perhaps from more acetate or butyrate produced in the rumen. However, ketosis was numerically decreased in another study (Shi et al., 2019). Further research is

needed to explain mechanisms, but more frequent meal consumption should lessen ruminal pH swings and help stabilize rumen fermentation.

Live yeast or yeast extract have been studied with young calves (Alugongo et al., 2017) to improve gut barrier function or perhaps for competitive exclusion of pathogens. However, these products also help reduce inflammation-based decreases in DMI or efficiency of nutrient usage, probably by priming the immune system to prepare for a subsequent challenge (Burdick Sanchez et al., 2020). Supplementation of live yeast (Bach et al., 2018) or yeast extract (Knoblock et al., 2019) in the dry period was assumed to predispose transition dairy cattle to inflammation and thereby dampen the subsequent inflammation response that spikes after calving. Resulting expression of genes in ruminal epithelium supported stronger barrier function. Consequently, live yeast (Al-Qaisi et al., 2020b) or yeast extract (Guo et al., 2022) also lessened leakage of bacterial endotoxin into blood and subsequent systemic immune response. Even heat stress (Al-Qaisi et al., 2020a) and mastitis (Vailati-Riboni et al., 2021) had dampened inflammatory responses when cows were fed yeast extract. Energy (Bertoni et al., 2015), carbohydrate (Kvidera et al., 2017), and AA (Coleman et al., 2020) are needed with an enhanced immune response and are diverted away from milk production, particularly with higher grain diets (Zebeli et al., 2015). Consequently, feed efficiency could be improved by yeast products under these conditions. More work is needed to explain variation, though, because the postabsorptive antioxidant capacity has not been consistently enhanced by feeding yeast extract (Knoblock et al., 2019; Sivinski et al., 2022).

Yeast products have some potential for young calves to resist inflammation-based diseases (Alugongo et al., 2017). As those authors noted, competitive exclusion of pathogens and improved gut barrier function need more research. Although outside of the scope of this paper, if these additives can survive the acidity of the abomasum and the detergent action of bile salts, they could potentially outcompete potential pathogens in the hindgut of lactating cows. Whether in the rumen or hindgut, inflammation and associated problems are being researched actively, particularly for transition dairy cows (Bradford et al., 2015; Zebeli et al., 2015).

ADDITIVES TO SUPPRESS METHANE

Numerous additives have been tested for strategies involved in methane abatement, which has involved teams of international scientists (Hristov et al., 2013) and the support of the US dairy industry (Knapp et al., 2014). Many methods to decrease methane have poten-

tial negative side effects (Beauchemin et al., 2020). For more expansive or detailed information on additives such as plant bioactives, readers are referred to recent reviews (Almeida et al., 2021; Honan et al., 2021). Some products are likely variable in composition or outcome and need further attention to variability in product and efficacy. For all additives, readers should exert caution when evaluating response variables that are ratios (e.g., methane:ECM and especially methane:DMI) because ratios can lead to problems with normality and interpretation (Løvendahl et al., 2018). Pragmatically, if an additive suppressed methane emission more than it depressed ECM or DMI, then a numerically increased ratio would likely not be well adopted at the farm level and also could result in no net benefit when factoring in more replacement animals needed to meet the same amount of milk produced; thus, life cycle analyses and other tools are needed to assess some additives. Animal selection for decreased methanogenesis might also select animals with reduced NDF degradability (Løvendahl et al., 2018) and, instead, should focus on decreasing methane while maintaining or improving feed efficiency (Manzanilla-Pech et al., 2021).

Lactate and Succinate as Fermentation Intermediates

A strong inverse relationship has been noted for the relative abundance of the family *Succinivibrionaceae* in the rumen and methane production (Mizrahi et al., 2021). Members of this group appear unable to use $H_2(aq)$ but instead are capnophilic, using CO_2 to capture $[2H]$ from upstream fermentation to sink to succinate, which is released and subsequently fermented by other microbes primarily to propionate. Similarly, Mizrahi et al. (2021) noted that animals with lower methane production appeared more positively associated with *Megasphaera* and *Coproccoccus* to convert lactate to propionate via the acrylate pathway. Their survey suggested *Coproccoccus* used $H_2(aq)$ when fermenting lactate and provided some evidence based on genomics, but we could find no empirical evidence to support that conclusion. In contrast with acrylate, some strains of *Selenomonas* can use H_2 to produce propionate via the succinate pathway (Asanuma et al., 1999).

Mizrahi et al. (2021) discussed the opportunity for faster passage rate to decrease methanogenesis. However, ruminal passage rate needed to be considered more holistically for dairy cattle (Firkins, 2021). For this discussion, our experience has been that readers too often misunderstand ruminal passage rate, which is a function of the ruminal pool size, not digestibility per se. In addition to an effect on methanogenesis, faster ruminal passage rate also should decrease fiber degradability

through faster washout of potentially degradable fiber, which would be exacerbated because of a less firmly packed ruminal mat and decreased rumination. Both decreased acetate:propionate and ruminal pH from such effects also will depress methanogenesis (Lana et al., 1998). Decreasing particle size could decrease digestibility but also could stimulate DMI to maintain ruminal fill (Beauchemin, 2018) and could actually increase intake of degraded fiber even with a lower digestibility. Therefore, processing forages enough to optimize rumination and intake of rumen-degraded fiber while minimizing the loss in degradability is critical to dairy production (Zebeli et al., 2012a). Passage rate depends on the type of forage. For example, legumes are more fragile and reduce in particle size more easily than grasses (Kammes and Allen, 2012). Finally, the interaction between forage particle size and inclusion rate of fibrous byproducts (Grant, 1997; Bradford and Mullins, 2012) further complicates the role of ruminal passage rate in the rumen. Combined with the challenge to even measure passage rate properly, its role in suppressing methanogenesis needs more research before it be demonstrated to play a systematic role in approaches to lessen methane without risking side effects.

Feeding Nitrate to Suppress Methanogenesis

Nitrate has a strong potential to lower methanogenesis in dairy cows (Honan et al., 2021) both as an alternative electron sink and by direct inhibition of methanogens by its intermediate, nitrite (Latham et al., 2016). Lactilytic strains of bacteria in the rumen also are likely important nitrate or nitrite reducers (mode 6 in Figure 1). The lactilytic *Selenomonas* strains represented only a small portion of the total *Selenomonas* isolates, yet the majority of those lactilytics also could reduce nitrate and nitrite (Yoshii et al., 2003). Surprisingly, little research has characterized if stimulation of lactate production (and therefore lactilytic populations) could prevent nitrite accumulation and whether or not ruminal ORP influences nitrate and nitrite reduction (Yang et al., 2016).

Practical nutritionists are well aware that a buildup of nitrite can lead to methemoglobin formation by animals consuming forages recently harvested during drought conditions or sorghum sudangrass following a frost. In contrast, this risk for feeding nitrate is lessened by ruminal adaptation through stepping up nitrate inclusion rates. Lee and Beauchemin (2014) concluded that nitrate would likely persist in methane abatement effectiveness over extended feeding periods. Meller et al. (2019) noted that decreased DMI dampened nitrate's effectiveness to decrease methane per unit of milk pro-

duced even though methane/DMI was improved (see previous discussion of ratios). Beef cattle fed diets with nitrate at ad libitum might naturally spread out their meals and exhibit more of a "nibbling" pattern of eating, whereas those that were restricted in intake consumed larger meals regardless of whether or not nitrate was added (Lee et al., 2015). Thus, energy demand could override palatability limitations, so researchers should be evaluating responses under ad libitum feeding conditions for a stronger inference. In contrast, Feng et al. (2020) suggested that increasing DMI dampened the antimethanogenic response of nitrate, whereas they suggested that use of slow-release nitrate sources fed to beef cattle (at lower DMI) might have enhanced the response. Because they did not directly assess if nitrate dose suppressed DMI, further work is needed to ascertain if slowing release or coating per se also help mask a potential palatability issue. If these issues could be worked out, the final product of nitrate reduction, ammonia, could be assimilated into bacterial protein in lower protein diets. The metabolizable energy spared by decreased methanogenesis associated with nitrate was not expected to offer much improvement in ruminant animal production (Lee and Beauchemin, 2014), and these conclusions have been noted for other methane-suppressing agents (Ungerfeld, 2018; Kelly et al., 2022).

Although there was considerable variation, a life cycle analysis documented moderate potential for nitrate to abate methanogenesis (Uddin et al., 2022), although assimilation of nitrate into microbial protein must be considered to prevent excess N excretion in urine. Sources of variation were related to many factors, which is why those authors recommended site-specific monitoring of methane. However, some of the variation might be related to the rumen microbiota. There is potential for using additives to stimulate lactilytic nitrate and nitrite reducers, especially those related to *Selenomonas*. Live yeast did not enhance this response (Meller et al., 2019), whereas a strain of *Paenibacillus* offers promise as a nitrite-using probiotic to be included when nitrate is fed (Latham et al., 2019). *Megasphaera elsdenii* encodes the *nirA* gene, which could help prevent nitrite accumulation; however, to our knowledge, no one has dosed this probiotic to animals fed nitrate. There likely are other nitrate or nitrite users in the rumen that have not been studied and could be used as nitrite-reducing probiotics.

3-Nitrooxypropanol

International efforts to evaluate the efficacy, persistence, and safety of 3-nitrooxypropanol (**3-NOP**) have largely been successful (Løvendahl et al., 2018). Dijk-

stra et al. (2018) noted that increasing dietary NDF decreased the efficacy of 3-NOP to suppress methanogenesis, whereas it was more efficacious with corn silage than grass silage (van Gastelen et al., 2022). When concentrate was decreased from approximately 50 to 30% of the diet (replacing corn and grass silages), persistence of methane abatement was diminished over the 98 d of the study (Schilde et al., 2021). Authors projected that increased forage increased H_2 production and therefore needed more 3-NOP dose as among the reasons for this response. Methylophilic methanogens appear to be less inhibited by 3-NOP (Pitta et al., 2022), so forage sources with more pectins or other precursors for methanol and methylamines also might need a higher dose. Three recent studies (Melgar et al., 2020a,b; 2021) documented that 3-NOP had minimal negative effects on palatability, DMI, and resultant milk quality while supporting its efficacy to suppress methanogenesis by approximately 25 to 30%. However, these latter 3 studies had divergent effects on ECM/DMI. Methane abatement increased with dose response in 1 study. The other 2 studies were longer term and supported persistence of 3-NOP over 15 wk. There was no 3-NOP treatment $\times$ time interaction in methane produced per unit of intake of digestible OM or NDF when 3-NOP was fed for 111 d (van Gastelen et al., 2020). Increased emission of H_2 or production of formate (when measured) could be underestimated if cows do not visit gas sampling units within the first hour after eating (van Gastelen et al., 2022), but H_2 emission was still be minor when measured when cows were in respiration chambers (van Gastelen et al., 2020) and does not seem enough to explain why 3-NOP efficacy was not consistent across forage source or amount.

Further research is needed using indirect calorimetry to better understand if and why additives such as 3-NOP that suppress methane do not improve ECM/DMI consistently. Researchers should consider metabolism of VFA and not just the greater heat of combustion of propionate versus acetate. In infusion studies, the acetate:propionate ratio was not associated with changes in energetics in nonlactating ruminants (van Houtert, 1993). Although 3-NOP decreased fiber digestibility in one study designed to assess energetics (Reynolds et al., 2014), it did not affect fiber digestibility in others (Melgar et al., 2020a; van Gastelen et al., 2020). Moreover, some potential differences in metabolism, such as increased mammary de novo synthesis of fatty acids, have been observed (Melgar et al., 2020a; 2021) and therefore could influence energy partitioning between adipose and mammary tissues. If decreasing methane increases propionate production (Schilde et al., 2021), then its role in insulin-mediated partitioning of energy should be considered in future studies.

Red Seaweed

Numerous research studies have been undertaken in the past decade to evaluate seaweed (macroalgae) as a methane suppressant. Most of the different species can be grouped into brown, green, or red seaweeds (Abbott et al., 2020). However, red seaweed, particularly genus *Asparagopsis* and especially one species, *Asparagopsis taxiformis*, are most widely researched (Almeida et al., 2021; Honan et al., 2021). It has multiple potential biologically active compounds (Glasson et al., 2022), most notably Br compounds and particularly bromoform. Other potential mechanisms to decrease methanogenesis include acting as an electron sink (Abbott et al., 2020). Those authors suggested a potential inhibition of bacteria and protozoa, which has not been widely studied, to our knowledge.

With respect to the animal, bromoform could have toxicity issues if fed at high doses (Muizelaar et al., 2021). In that study, cows had 1 to 2 h to consume a concentrate containing the *A. taxiformis* product, likely exacerbating a toxicity (Glasson et al., 2022). Those authors suggested that, with moderate feeding to adapted animals, milk consumers should have very minimal risk. Glasson et al. (2022) further explained that many of the compounds are broken down in the rumen, but, when fed at high doses, halogenated compounds can be excreted in urine or milk without accumulating in tissues. In addition to Br, elevated concentrations of I can occur in milk from cows fed seaweed (Lean et al., 2021; Stefenoni et al., 2021). Pooling of milk with that of cows not fed seaweed was projected to limit any residue issues (Stefenoni et al., 2021). In contrast, this dilution approach would require limited adoption to be effective. Milk I concentration in various international systems varies widely because of a variety of factors (van der Reijden et al., 2017). Therefore, more research is needed across a variety of feeding systems to minimize risk of excess I in milk for consumers.

Although improved milk or milkfat production without a depression in DMI was noted in a meta-analysis (Lean et al., 2021), potential depressions in DMI by dairy cattle (Stefenoni et al., 2021) also should be explored. The large range in responses (Lean et al., 2021) suggest potential interactions with animal or dietary variables, although decay of the active ingredient with increased storage (Stefenoni et al., 2021) also needs further attention. If the various details about decay and milk residue can be worked out, red seaweed offers strong potential for suppressed methanogenesis (Kinley et al., 2020; Almeida et al., 2021). Even so, logistical details need to be worked out to maintain sustainable seaweed farming (Abbott et al., 2020) and to address consumer concerns about numerous issues such as the

marine and land environments, product regulation, and animal and human safety (Vijn et al., 2020).

ADDITIVES POTENTIALLY IMPROVING GUT HEALTH OR FEED EFFICIENCY

Lactate-Producing Probiotics

There is some potential to gain a benefit by administering lactate-producing probiotics (McAllister et al., 2011). Those authors explained that there are strain variations likely behind the variable responses in the published literature. There is some potential for lactate producers to help maintain a healthy hindgut by inhibiting potential pathogens such as *Escherichia coli* 0157 (Doyle et al., 2019). Those authors also discussed a potential role for probiotic lactic acid bacteria to decrease methanogenesis. The mechanistic evidence seems sound, although there is the caveat that these probiotics probably should only be considered when there is not a SARA risk. A lactate-producing probiotic is probably more effective to suppress methanogenesis when combined with a lactolytic probiotic (Philippeau et al., 2017), which has been noted for increased populations of both commensal lactate producers and lactilytics in naturally low methane-emitting animals (Ungerfeld, 2020). Ferraretto and Shaver (2015) noted no major differences in lactation performance when a probiotic combining a lactate producer and consumer was added to the TMR. They discussed some studies in which ECM efficiency was improved. However, they speculated that high starch availability might have attenuated their study's potential response. More research is needed to explain these sources of variation among studies (Jeyanathan et al., 2019; Honan et al., 2021).

Ionophores

Ionophores have been very well studied for their improvement in feed efficiency. Among other modes of action, they are thought to increase propionate production while inhibiting lactate-producing bacteria and other bacteria that enhance proteolysis and deamination (McAllister et al., 2011; Firkins and Yu, 2015). We will limit our discussion to a few points clarifying misconceptions in the scientific literature. After its US legalization for feeding in lactating dairy cattle in 2004, monensin was noted to tilt biohydrogenation toward the *trans*-10 C18:1 pathway in cows fed diets high in fermentable starch and unsaturated fats (particularly with free oils such as those in distillers grains), but most of these types of situations were worked out thereafter (Firkins et al., 2007). Increasing dosage for beef cattle (on a dose/DMI basis) relative to dairy likely de-

creases acetate:propionate and methanogenesis (Ranga Niroshan Appuhamy et al., 2013). The slight decrease in DMI (part of the feed efficiency response) probably also has a role in suppression of methanogenesis (Patra et al., 2017), particularly when expressed per unit of ECM (Hristov et al., 2013). Increasing propionate has obvious ramifications for postabsorptive function (e.g., decreasing risk for ketosis), but monensin's role might extend beyond propionate (Mullins et al., 2012). Those authors noted that monensin increased feeding frequency, which could help stabilize ruminal fermentation.

Ionophores have many benefits related to stabilization of ruminal fermentation and to improved nitrogen and energetic efficiency (Firkins and Yu, 2015). Microbial adaptation explains why major inhibitions of gram-positive bacteria in the landmark pure culture studies are lessened in the rumen of adapted animals (Schären et al., 2017). Protozoa also adapt to monensin, although their generation time might be lengthened (Ye et al., 2018). Many of the methanogens associated with protozoa adjust to alternative bacterial H₂ producers if protozoa are suppressed by rumen modifiers, and protozoa certainly have benefits compared with disadvantages related to ruminal N recycling and enhanced methanogenesis (Firkins and Yu, 2015; Firkins et al., 2020). Hence, many protozoal suppression efforts had side effects such as depressed DMI or lower NDF degradability (Tapio et al., 2017). In fact, many other feed additives suffer the same variability and potential for adaptation such that suppression of protozoal counts has an uncertain role in methanogenesis abatement strategies (Hristov et al., 2013).

Monensin inclusion in dairy diets is not been consistently recommended as a methane abatement strategy because of microbial adaptation (Almeida et al., 2021). Therefore, persistence for lowering methanogenesis through improved feed efficiency has been questioned based on mixed responses in studies (Knapp et al., 2014). However, some of these studies have used methane measurement approaches that lack accuracy or at least precision (Odongo et al., 2007). Papers that did not address critical issues (e.g., calibration, repeatability) related to methane measurement (Hristov et al., 2018) should be reassessed for their potential merit.

Branched-Chain VFA

We have known since the 1950s that some prominent cellulolytic bacteria in the rumen require branched-chain volatile fatty acids (BCVFA), which are produced only when microbes deaminate or transaminate the amino group and decarboxylate the keto acid from the branched-chain AA in RDP (Firkins, 2021). The BCVFA can be reductively carboxylated back to

branched-chain AA or elongated to branched-chain fatty acids needed in the membranes of anaerobic bacteria (top left of Figure 1). Andries et al. (1987) noted 4 to 8% increase in milk production from feeding IsoPlus. Some studies varied the concentration of the BCVFA plus valerate, but most studies had improved feed efficiency (Papas et al., 1984; Peirce-Sandner et al., 1985), although this response depended on DIM and other factors. Many of these studies were with cows with lower propensity for milk production and not fed TMR. With increasing milk yield and increasing pressure to limit N excretion in waste by decreasing RDP, these isoacids are being reinvestigated.

The degradability of NDF generally increased 3 to 5% unit in batch cultures (Roman-Garcia et al., 2021a) and continuous culture (Roman-Garcia et al., 2021b). Microbial N flow was not affected, but it increased by 7% in another study that also tended to increase NDF degradability (Mitchell, 2022). In these diets, RDP was moderate (9.5+%), but urea was supplied in the buffer to prevent an ammonia deficiency. When RDP apparently was not limiting in lactating dairy cows, no benefit in NDF digestibility was noted (Copelin et al., 2021), but the BCVFA helped prevent milkfat depression apparently by stimulating de novo synthesis in the mammary gland (Lee et al., 2021). Thus, there could be both ruminal and postruminal benefits to feeding isoacids.

Supplemental valine (degraded to isobutyrate) stimulated milk production (Hultquist and Casper, 2016), potentially through a postabsorptive response as evidenced by an increase in blood thyroxine concentration. Supplementing all 3 BCVFA increased DMI and production of milk components (Wang et al., 2019). Increased growth hormone and other related plasma variables suggested a potential shift in partitioning of nutrients to the mammary gland (no change in BW gain was noted). Increased expression of genes related to mammary de novo synthesis was explained by increased fiber digestibility and acetate production in the rumen (Liu et al., 2018). Both of these latter 2 studies had all 3 BCVFA in a 1:1:1 weight ratio, used only multiparous cows, and started both trials at 65 DIM. In contrast, feeding only isobutyrate at 3 concentrations to multiparous cows starting at 148 DIM had quadratic responses in milk and milk fat coinciding with increasing energy partitioned to BW (Liu et al., 2009). Mitchell et al. (2020) noted that feed efficiency improved by over 5% when isobutyrate and 2-methylbutyrate were supplemented to Jersey cows.

Plant Components and Extracts

Many studies have explored various plant components and extracts that inhibit (1) methanogens to

suppress enteric methane production and (2) decrease protein wastage by inhibiting ruminal protozoa and the bacteria that rely on AA as energy sources (hyperammonia producers; Cobellis et al., 2016; Patra et al., 2017; Hartinger et al., 2018). Much of the efficacy is projected from interactions with susceptible microbes' cell walls (Benchaar and Greathead, 2011). Similar to microbial adaptation to monensin, microbes appear to adapt to many of these plant bioactive compounds by changing the architecture of their cell wall. More longer-term in vivo studies are needed to demonstrate the efficacy of plant bioactives in reducing methanogenesis (Beauchemin et al., 2020).

Each class of plant bioactive compounds has various chemicals involved in their modes of action. Tannins have had positive and negative results to inhibit undesirable bacteria and protozoa (Hartinger et al., 2018). However, those microbes also probably can adapt by modifying their cell wall. Tannins also can reduce protein degradability in the rumen, but the advantage would be lost if they also decrease the intestinal digestibility of RUP. Hartinger et al. (2018) noted variable responses (e.g., to inhibit protozoa) from feeding saponins, including that they also sometimes have decreased DMI. Essential oils have received considerable study (Cobellis et al., 2016) and offer some potential in dairy nutrition. Ye et al. (2018) hypothesized that combining monensin and an essential oil would inhibit protozoa and suppress methane in continuous culture, whereas the opposite result occurred. Protozoal suppression must be enough to inhibit their numbers and negative activities but not so much to limit overall microbial function; this middle ground is hard to find. More studies on combinations of bioactives are needed and for longer time to account for potential microbial adaptation.

Combinations of plant bioactives might limit the general inhibition by each individual compound while gaining more consistency and efficacy (Beauchemin et al., 2020). Saponins and tannins have been explored with mixed benefit (Almeida et al., 2021; Honan et al., 2021). Mixes of compounds might offer some potential gain in consistency not just by combining different compounds, but also by minimizing risk of any single compound having lower concentration of active ingredient than expected. Belanche et al. (2020) reported that a combination of compounds in one commercial product maintained persistency and, in fact, documented a greater response with increasing time of evaluation. Feed efficiency was improved by about 4%, and methane production per unit of energy- and protein-corrected milk was decreased by about 10%. Not all studies report significant inhibition of methane production or methane intensity (Honan et al., 2021). Hence,

more research is needed to evaluate combinations of additives to improve consistency of effectiveness (i.e., variability of active ingredients in individual sources, decay of activity over time in storage, or adaptation by the rumen microbiota) before we can understand and predict dietary situations most amenable to the potential benefits of plant bioactives.

CONCLUSIONS

Rumen modifiers should not be used to substitute for good feeding management. They can reduce variability among animals to improve feed efficiency and decrease rumen methane production with less risk for decreased milk or milk fat production. Many of the additives discussed herein have a mechanism involving fermentation pathways, particularly those using lactate. However, increased production of lactate as an intermediate also should decrease methane per unit of ECM so long as lactate accumulation is minimized in the process. In addition to stimulating lactate usage, yeast products appear to decrease inflammation in the gut and lessen partitioning of nutrients away from milk. Monensin is fed in many countries to increase feed efficiency, but some of the response appears to be more than just increasing propionate, so it should have further study when combined with other rumen modifiers. Many compounds have been tested to suppress methanogenesis, with 3-NOP and red seaweed having the largest response. In contrast with 3-NOP, which has a recognized double mode of inhibition (Duin et al., 2016), other additives that competitively inhibit methanogenesis have had more variable persistency, suggesting rumen microbial adaptation or other limiting issues such as decreased DMI (Patra et al., 2017). Those authors suggested combinations of additives spanning multiple mechanisms might hold promise. When using rumen bioactives, we need to understand their modes of action to better justify their usage but also to understand what variables can limit their benefits. Finally, any additive that is not generally recognized as safe will need to pass safety tests appropriate for the respective country in which they are marketed.

ACKNOWLEDGMENTS

Research was jointly supported by state and federal funds appropriated to the Ohio Agricultural Research and Development Center, The Ohio State University (Columbus, OH). A prior proceedings was provided for the 2021 Tri-State Dairy Nutrition Conference, Fort Wayne, Indiana. No human or animal subjects were used, so this analysis did not require approval by an Institutional Animal Care and Use Committee or Insti-

tutional Review Board. J. L. Firkins has current funding from Zinpro Inc. to study BCVFA. K. E. Mitchell currently works for Elanco Animal Health, which has a financial interest in monensin and 3-NOP. The authors have not stated any other conflicts of interest.

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